

Habit Arena — Product Requirements Document (PRD)

Based on a walkthrough of a live reference product's public site and its original team's project write-up — used only as inspiration for the concept, not for the name or branding.

1. Introduction

Habit Arena isn't really a habit tracker — it's a social competition engine pointed at your personal goals. Instead of a solo streak counter that quietly resets after one bad week, Habit Arena puts a small group of friends, a shared goal, a deadline, and (optionally) something to lose, all inside one private "room," then ranks everyone on a live leaderboard until the room closes and a prize pot gets split among the top finishers.

Alongside that, Habit Arena also lets a user track a habit entirely solo — no room, no rivals, just a personal streak. But the rivalry mechanic — competing against real people with real stakes — is the app's main point. Solo tracking is a supporting feature, not the pitch.

Your mini-project is to build this concept as a single responsive web app that works well on both a laptop and Android. The app is free to use for everyone.

User Personas

- **The Competitor.** Wants an audience and something at stake. Joins or creates rooms, checks the leaderboard daily, is motivated by rank and the pot. This is the primary user the product is designed around.
- **The Quiet Starter.** Wants to build a habit but isn't ready to be watched yet — maybe the habit is personal, or they just want to test their own consistency first. Uses solo mode, may graduate into a room later once they trust the habit will stick.
- **The Group Organizer.** The friend who creates the room, sets the topic/duration/entry fee, and shares the invite link to get a group moving together. Cares about the room being easy to set up and easy to get others into quickly.

Non-Functional Requirements

- **Performance:** Leaderboard and check-in updates should feel near-instant (no full page reloads needed to see an updated rank).

- **Availability:** As a mini-project, uptime guarantees aren't required, but the app shouldn't lose a user's streak or check-in data on a crash or refresh.
- **Security:** Standard authentication (hashed passwords or an auth provider), and room access restricted to invited members only — no public browsing of a room's data.
- **Data integrity:** A user can log at most one check-in per day per habit/room; the payout split at the end of a room must add up to the full pot with no double-counting.
- **Responsiveness:** One codebase, adapting cleanly between a mobile (Android) layout and a wider desktop/laptop layout — not two separate apps.

Success Metrics

- **Room completion rate** — % of created rooms that run to their full duration rather than being abandoned early.
- **Daily check-in rate** — % of active participants who check in on a given day within a live room.
- **Solo-to-room conversion** — % of solo-mode users who go on to join or create a room (a signal that the rivalry mechanic is pulling people in, as intended).
- **Retention** — % of users who return to check in again the following week.

2. The Problem It's Solving

Most self-improvement apps fail for an unglamorous reason: nothing actually happens when you quit. You let a streak lapse on a to-do app and nobody notices — including, eventually, you. The core insight here is blunt: the issue was never a lack of willpower, it was a lack of anyone to compete against. Left alone, most goals fade out quietly, because there's no audience and nothing at stake.

Habit Arena attacks that gap from three directions at once:

- **Social comparison.** A lot of people find it easy to break a promise to themselves but noticeably harder to break one when a friend is watching and keeping score.
- **Visibility as accountability.** A habit nobody else can see is one you can quietly abandon. Once your progress is visible to a small group, letting it slide stops being a private, cost-free choice.
- **Reframing effort as play.** Turning daily progress into rankings and a pot to win makes a chore feel closer to a game you're trying not to lose — a very different trigger than "self-improvement."

This lines up with an established idea in behavioral design called a *commitment device*: you voluntarily attach a cost to failing, so quitting becomes

expensive rather than just quietly disappointing. Layering real stakes and peer visibility on top of an otherwise ordinary tracker is what changes how people treat their own goals — that's the entire thesis of the product.

The solo mode exists for the (real) cases where a person doesn't want an audience yet, or the habit is too personal to share — but it's deliberately the lighter-weight option next to the rivalry mode, not a replacement for it.

3. Objective of Your Mini-Project

Build a platform where:

1. A user can track a habit **solo** — define it, log daily progress, keep a personal streak, no room or rivals involved.
2. A small, closed group of people can:
 - Define a goal or habit as a time-boxed "challenge"
 - Invite only the people they choose into a private room
 - Log progress daily, building a streak inside that room
 - See everyone's standing on a live leaderboard for the room

- (Optionally) put a small amount of virtual points into a shared pot, redistributed to the top performers once the challenge ends

The rivalry/room mode is the core of the product and where most of the design effort should go. Solo mode should reuse the same daily check-in and streak logic under the hood — it's the same core mechanic, just without a room wrapped around it.

4. What the Reference Concept Actually Involves — A Closer Look

A few things stand out once you look past the marketing:

It's topic-agnostic by design. A room's topic is free text — gym consistency, no late-night scrolling, no contact with an ex, a daily cold-shower streak, a coding-practice leaderboard, whatever. The engine underneath doesn't care what the goal is — it only needs a way for someone to report "yes, I did it today." That's the actual core mechanic to build: a generic daily check-in attached to an arbitrary, user-defined goal, not anything fitness- or study-specific. This is exactly the mechanic solo mode reuses.

"Winner takes all" isn't literally how it should work. A pure winner-takes-all payout is a harsh sell. A tiered payout — first place gets the largest share, but second

and third still walk away with something — is friendlier and worth building instead: it keeps more people engaged for the whole duration, instead of checking out the moment they fall out of first place.

The full concept can be bigger than the core loop.

Beyond create-room → invite → compete, a fuller version could run a persistent social layer outside any single room — a friends list, direct messages, win-streak tracking that follows a user across rooms — plus live chat inside an active room. That's a lot of extra surface area, and it's the first thing to cut given limited time (more on scoping in section 6).

Visual identity matters. A bold, high-contrast, game-like visual style — thick outlines, hard drop shadows, no soft gradients, competitive/needling copywriting — does real work here. The same feature set with a neutral, corporate skin would land very differently. You're free to build your own visual identity for Habit Arena; nothing about the original's branding, name, or mascot should carry over.

A public waitlist/landing page is a separate concern from the product. That's pre-launch marketing for collecting sign-ups before an app exists — it has nothing to do with the core system you're demonstrating, and you don't need to build anything like it.

5. Core User Flows

Two flows, sharing the same underlying check-in/streak engine:

Solo Flow

1. Create a personal habit (name, optional target frequency).
2. Log a daily check-in.
3. See your own streak and simple stats — no room, no rivals, no pot.

Rivalry Flow (main flow)

1. **Create a Room.** The creator fills in a topic (free text), a duration in days, a maximum number of participants, an entry fee (virtual points), and a start date. The total pot — entry fee × number of participants — is shown up front, so everyone knows what's at stake before joining.
2. **Invite Rivals.** Creating a room generates a unique, shareable invite link, meant to be dropped into an existing group chat. Invitees have a short window to accept before the invite expires, which keeps rooms from sitting half-full indefinitely and adds a little urgency to joining.
3. **Compete & Win.** Once a room is live, each participant logs progress daily. That builds a personal streak and feeds a live leaderboard for

the room. When the set number of days runs out, standings lock in and the pot is distributed by rank.

6. Feature Breakdown: Core vs. Future Scope

Deliberately separating what's essential to prove the concept from extra polish is worth doing early — trying to build the entire social platform in one pass is how mini-projects run out of time.

Core (build this first)	Future Scope (mention, don't build yet)
Account creation / login	Persistent social layer — friends list, DMs, cross-room stats
Solo habit tracking — define habit, daily check-in, personal streak	Live chat inside an active room
Create room: topic, days, max participants, entry fee, start date	AI-generated daily prompts or quiz-style tasks
Shareable invite link with an acceptance deadline	Custom mascot/avatar that levels up with points earned

Core (build this first)	Future Scope (mention, don't build yet)
Daily check-in / progress log (one per user per day, shared logic for solo and room)	Public matchmaking / leagues beyond invite-only rooms
Per-user streak counter within a room	Fitness-tracker/wearable integration to auto-verify habits
Live per-room leaderboard	
End-of-challenge results screen with rank-based payout	
Simple personal stats view (streaks, active rooms, today's rank)	

7. Screens You'll Likely Need

- Login / Sign-up
- Dashboard — active rooms, solo habits, personal streaks, quick stats
- Create Solo Habit form

- Create Room form
- Room invite / "waiting for others to join" screen
- Room detail screen — leaderboard, your streak, today's check-in action
- Solo habit detail screen — streak, check-in action, history
- Final results screen — locked standings and payout breakdown
- Profile — badges, streak history, points balance

8. Design Decisions That Are Genuinely Yours to Make

These are legitimate open decisions for your report rather than things you'd be "getting wrong" by choosing differently:

- **How is score calculated, and how does it map to payout?** You'll need to define both how score accumulates day to day in a room and how it converts into a share of the pot.
- **Self-reported or verified check-ins?** Honor-system logging is far simpler to build; verification (timestamps, proof uploads, device integration) is more robust but adds real complexity. This applies to both solo and room check-ins.
- **Mandatory or optional stakes?** You could require every room to have an entry fee, or let it default to

zero for a "casual" no-stakes room.

- **What's the payout curve?** A steep winner-take-most split rewards competitiveness; a gentler tiered split (something for anyone above a minimum threshold) keeps mid-pack participants motivated instead of checking out early. Worth a deliberate paragraph in your report rather than an afterthought.
- **What happens when someone drops out or misses check-ins?** Forfeit the stake entirely, forfeit it partially, or just fall in rank with no extra penalty — each is defensible, but pick one and state your reasoning.
- **Tie-breaking rule** for identical leaderboard scores.
- **Does solo mode ever feed into anything competitive?** For example, should a personal streak be shown on a profile as a badge/stat, even though it has no stakes or rivals — purely as a "consistency" indicator? This is optional, but worth deciding.

One practical scoping note: sticking with a virtual, non-redeemable points currency (not real money or crypto) is the right call for a college mini-project — real-money stakes pull in payment processing and gambling-adjacent regulation that's well outside the scope of what you're trying to demonstrate.

The app itself is free to use. There's no charge to sign up, create a room, join a room, or use solo mode. The entry-fee/pot system runs entirely on virtual points with no real-money component anywhere — so "free to use" and "stakes exist in a room" aren't in tension; nobody ever pays real money, win or lose.

9. Why This Is a Well-Scoped Mini-Project

The core loop touches a genuinely useful spread of concepts: authentication, relational data modeling (users, rooms, participants, and daily logs all reference each other), a ranking/scoring engine, near-real-time leaderboard updates, and simple transactional logic for splitting a pot without double-counting anyone. Adding solo mode is low additional cost, since it reuses the same check-in/streak engine — it mainly adds a "habit with no room" data path. That's substantial enough to be a real project without needing the full social platform to prove the idea works. Since you're building one responsive app for both laptop and Android rather than two separate apps, the main UX puzzle is adapting a tall, card-stacked mobile layout into something that also reads well on a wider desktop screen — worth sketching both breakpoints before you start building screens. It's also a solid, demoable piece to have alongside

your other work when internship applications come around.

10. Conclusion

Strip away the branding, and Habit Arena is a small, elegant idea: take the private, easy-to-abandon discipline of habit-building and make it social, visible, and time-boxed, with just enough at stake that quitting has a cost — with a lighter solo mode alongside it for habits someone isn't ready to compete on yet. The core loop remains: create-room → invite → daily-log → leaderboard → payout, plus a simple solo-tracking path using the same underlying mechanic. Everything else (chat, DMs, matchmaking, custom mascots) is genuinely optional polish, better listed as future scope than built into a first version.